1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In MIBC, MSH6-deficient mismatch repair perturbations drive high TMB but bias neoantigen landscapes toward subclonal and/or low-quality neoantigens, limiting durable T cell–mediated tumor clearance despite high TIL density. Therefore, integrating histologic TIL density with MSH6 status and TMB will predict immunotherapy response more accurately than TIL or TMB alone by capturing the balance between immune infiltration and neoantigen clonality/quality.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, TMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6, HRD, TMB): why do some TIL-rich MIBC tumors fail to benefit from immune therapies despite high mutation loads? 2) Tier 1 paths used: I built on the MSH6 → prognostic factor path and the TMB (immunotherapy predictor) path because they directly connect DNA-repair status to neoantigen generation and immune recognition; these provide the clearest mechanistic axis to explain variable immune responsiveness.

TMB: TMB): why do some TIL-rich MIBC tumors fail to benefit from immune therapies despite high mutation loads? 2) Tier 1 paths used: I built on the MSH6 → prognostic factor path and the TMB (immunotherapy predictor) path because they directly connect DNA-repair status to neoantigen generation and immune recognition; these provide the clearest mechanistic axis to explain variable immune responsiveness.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	HRD_score	Supporting modifier from knowledge graph

Secondary	TP53	Supporting modifier from knowledge graph
Secondary	MSI_phenotype	Supporting modifier from knowledge graph
Secondary	urine_DNA_methylation_biomarkers	Supporting modifier from knowledge graph
Secondary	AI-based_biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- HRD_score: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.